

A SHORT GUIDE

BASE-12 KAKTOVIK

CALCULATOR

Dozenal arithmetic with Kaktovik glyphs. This booklet shows the gestures that change the keypad, the numeral set, and the scientific layout — plus how long formulas stay on the screen.



Kaktovik · left ops



Arabic · left ops



Kaktovik · right keys

WHAT YOU WILL LEARN

- Hold the display to switch Kaktovik glyphs and Arabic 0-9 A B.
- Swipe the display upward to move the number keys to the right.
- Tap BASIC / ADV for scientific keys, then hold dual labels for the second function.
- Tap RAD to use dozenal turns (12T). Formulas compute only when you press =.
- Long equations stay inside the screen; drag history left or right to read it.

THE SCREEN



The black panel is the only place formulas appear. Nothing is allowed to spill off the calculator body.

History line

The faint row at the top is the last finished equation. If it is too long, drag it sideways with your thumb.

Live equation

The middle row is what you are building. As it grows, it scrolls so the newest symbols stay in view on the right.

Result

The large glyphs are the number you just typed — or the answer after you press =.

RAD / ₁₂T

Tap this label to switch angle units. RAD is radians. ₁₂T is dozenal turns: a full circle is 10 in base 12 (twelve units).

o sub ₁₂

The small Arabic readout is the same value written with 0-9 A B and a base-12 subscript.

SWAP TO ARABIC NUMERALS

The default keypad uses Kaktovik stroke-glyphs. Hold the black display for about one second to switch the digit keys — and the big result — to ordinary 0-9 plus A and B for ten and eleven. Hold the display again to return to Kaktovik.



Kaktovik glyphs



Arabic 0-9 A B

HOW TO SWITCH

- Press and hold anywhere on the black display for about 1.2 seconds. A short tap will not do it.
- The digit keys change immediately. A and B stay as the extra dozenal digits for ten and eleven.
- In Arabic mode the display also shows a small decimal equivalent, written $= n$ with a ₁₀ subscript.
- Hold the display again to go back to Kaktovik. The value you were working on is kept.

A and B are still base-12 digits, not hexadecimal. 10 on this calculator is twelve, written 10 in base 12.

MOVE THE NUMBER KEYS TO THE RIGHT

A quick upward flick on the display mirrors the keypad. Operators slide to the left; the dozenal digits sit on the right, under your thumb if you hold the phone in your right hand. Flick up again to restore the original layout.



Default — operators on the right



Flipped — number keys on the right

HOW TO FLIP

- Touch the display and swipe upward at least a finger-width, in under about half a second.
- A slow drag or a sideways drag will not flip the pad — sideways is reserved for reading long formulas.
- Flip works in BASIC and ADV, and it stacks with Arabic numerals. You can use both at once.
- The equals key travels with the operators so the colored column stays together.

BASIC AND ADV

BASIC is four columns: digits, the four arithmetic signs, parentheses, reciprocal, and memory add. ADV opens a six-column scientific pad. Tap the BASIC / ADV segment control under the display.



ADV — scientific column on the right



ADV + flip — numbers on the right

ADV KEYS

mod / %

Tap for remainder (modulo). Hold for percent of $_{144}$, the dozenal hundred.

sqrt

Prefix square root. Write $\text{sqrt } 2$ or $\text{sqrt } (2 + 3)$.

$\wedge$ / e^x

Tap for power. Hold for e to the power of the next value.

Tau / π

Tap inserts tau (a full turn). Hold inserts pi.

sin cos tan

Tap the function. Hold sin or cos for asin / acos.

log₁₂ / ln

Tap for log base 12. Hold for natural log.

M+ M- MR/MC

Memory add, subtract, recall. Hold MR/MC to clear memory.

HOLDS, TAPS, AND MEMORY

Keys with two labels use a short tap for the large label and a hold of about half a second for the small one. The same idea applies to DEL / AC and to MR / MC.

DEL

Deletes the last digit or token.

Hold DEL (AC)

Clears the current expression to zero.

+/-

Changes the sign of the number you are typing, or of -pi / -tau / -e.

1/x

Reciprocal of the current value or of a finished group.

M+

Adds the displayed value to memory. Does nothing while the display reads Error.

M-

Subtracts the displayed value from memory (ADV).

MR

Places memory into the expression. Hold MC to wipe memory.

()

Group work. Closing a parenthesis does not compute the whole line.

=

Evaluates the expression on the middle row. This is the only time a long formula is solved.

RAD

Tap the RAD label to toggle 12T. Trigonometry follows the flag.

If the display shows Error, only a new digit, DEL, or AC will leave that state. Operators, memory plus/minus, and functions stay inert so a failed line cannot turn into a quiet zero.

LONG EQUATIONS AND EVERYDAY USE

Build the line on the middle row the way you would write it: $2 \times (\sqrt{2})$, then press $=$. The large glyphs stay on the last number you typed until equals. After $=$ they become the answer, written in the numeral set you chose.

WHEN A LINE IS TOO LONG

- The live equation never draws outside the black glass. The left side slides out of frame; the right side — what you just typed — stays visible.
- Drag that middle row left to review the beginning, or let it snap back to the end as you keep typing.
- The faint history row is the last completed line. Drag it left or right the same way. A vertical flick on the display still flips the keypad; a horizontal drag will not.

A FEW WORKING RULES

- All arithmetic is base 12. $10 \times 10 = 100$, which is one gross. A and B are ten and eleven.
- Percent is per 144, not per 100. That matches a dozenal hundreded.
- Install it as an app from the browser share sheet (Add to Home Screen). The icon is the wooden abacus; it will open full-screen.
- Arabic numerals, flipped keypad, and ADV can all be on at the same time. None of them change the math — only how the keys look and where they sit.



Arabic + right-hand keys

ADV + Arabic numerals